

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2297

West Barlett 138 kV
April 16, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2297. The Interconnection Customer proposes a 87.9 MW battery energy storage facility interconnecting at the West Barlett 138 kV substation in the Maryland area. The request was received on March 13, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 670 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,931,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ritzville / Carlotta 138 KV line
Contingency: Air Force Base 345-138 kV Transformer Bank
Loading: 117.5% (Rating: 276 MVA)
DFAX: 20.7%
Pre-project loading: 110.9%
2. Facility: McCormick Substation / Dranesville 138 KV
Contingency: Dranesville / McCormick Substation 138 KV Line 2
Loading: 100.8% (Rating: 393 MVA)
DFAX: 5.2%
Pre-project loading: 99.6%
3. Facility: McCormick Substation / Dranesville 138 KV
Contingency: McCormick / Dranesville 138 KV
Loading: 100.8% (Rating: 393 MVA)
DFAX: 5.2%
Pre-project loading: 99.6%
4. Facility: Dekalb / Robison Park Substation 230 KV
Contingency: Dekalb / Jones Chapel 500 KV
Loading: 100.1% (Rating: 3,906 MVA)
DFAX: 7.4%
Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carlotta / Ritzville 138 KV	Rebuild line with higher-capacity conductor	\$4,356,000	48% (\$2,076,000)
Dranesville / McCormick 138 KV Cir. 1	Reconductor line	\$1,044,000	38% (\$396,000)
Mccormick Substation / Dranesville 138 KV	Reconductor line	\$1,100,000	38% (\$417,000)
Dekalb / Robison Park Substation 230 KV	Reconductor line	\$11,268,000	0% (\$42,000)
	Total allocated to project		\$2,931,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.